or `next.revalidatePath()` to specify how often Next.js should check for new data. If a request comes in after the specified time, Next.js will serve the old page while it generates a new one in the background.

PERFORMANCE OPTIMIZATIONS

Next.js offers a few components and configuration parameters that allow fine-grained control of caching and efficiency:

- *Image and font optimization*—The Next.js `<Image />` component automatically optimizes images, serving them in modern formats like WebP or AVIF and resizing them for different devices. Additionally, the `@next/font` module automatically optimizes fonts, removing unused font files and ensuring they are loaded efficiently.
- *Lazy loading*—The `@next/dynamic` module allows you to lazy load components, which means a component and its dependencies are only loaded when they are needed. This is great for large components that are not immediately visible on the page, significantly reducing the initial page load time.
- *Caching with the fetch API*—Next.js extends the native fetch API to provide a powerful caching mechanism. You can control the caching behavior of individual fetch requests using options like `cache: 'no-store'` for dynamic data or `next: { revalidate: 3600 }` to set a revalidation time for static data.

You can explore more detailed features in the official Next.js documentation at https://nextjs.org/docs.

Now we're ready to look at the Vercel AI SDK and see how we can include features like streaming support, backpropagation, and cancellation into our application. These functionalities become particularly relevant when building real-time, interactive conversational interfaces.

Summary

- Building generative AI applications involves several crucial steps. You need to plan a design, the key features it will offer, and how it will be delivered to your users. Then the implementation needs to use all the best practices of UX, security, and testing to ensure it works as intended.
- The planning and design phase focuses on defining the desired features and functionalities of your application. Selecting the right tools to fulfill those needs is crucial for success.
- The simplest conversational chat web app features a way to accept user input (via a message box) and send this to the backend.
- When building user interfaces for generative AI applications, UX becomes paramount, especially when dealing with lengthy LLM responses.
- Security is a top priority for the backend, ensuring the safe handling of user data and application logic. Additionally, performance optimization becomes crucial